THE INVARIANT

Wave Recovery — Formal Appendix

A rigorous treatment of the holomorphic lifting,
the Bernoulli constant, and the surface recovery formula

The central result is non-monotonic: reconstruction error improves as N increases up to an optimal order, then diverges as $\exp(nkd)$ amplifies floating-point noise back through the fluid column. This appendix works through why.

The method presented here follows D. Clamond, *New exact relations for easy recovery of steady wave profiles from bottom pressure measurements* (2013). An earlier procedure due to D. Clamond and A. Constantin (2012) is treated in the complementary questions. Code is available at github.com/theinvariantlab/the-invariant.

In this appendix

1 What is a Stokes wave? 2

2 Why does the pressure at the bed determine $\mathcal{B}$? 2

3 What is $\mathcal{D}_s$ and where does it come from? 3

4 How is η extracted from $\mathcal{D}_s$? 4

5 How is B determined? 5

6 What breaks at high N? 5

Complementary questions 6

The results in this appendix rest on deeper analytical tools: maximum principles for elliptic PDEs, the hodograph conformal transformation, and Banach fixed-point theory. These are developed in the original manuscript.

1. What is a Stokes wave?

Setup. We consider a two-dimensional periodic wave in a cross-section perpendicular to the crest line, with coordinates (x, y), the y-axis pointing vertically upward. The fluid domain is

$$\Omega = \{(x, y) \in \mathbb{R}^2 : -d < y < \eta(x)\},$$

with $y = -d$ the flat impermeable bed and $y = \eta(x)$ the unknown free surface. The flow is irrotational and inviscid. We write u, v for the horizontal and vertical velocity components, p for the normalised pressure, and g for gravitational acceleration.

Definition 1.1 (Stokes wave)**.** A Stokes wave is an irrotational smooth solution to the governing equations for which there exists a period $L > 0$ and wave speed c such that η, u, v and p have period L in x. There is a single crest and trough per period; the profile is decreasing from crest to trough. The functions η, u, p are symmetric and v is antisymmetric about the crest line.

Passing to the moving frame eliminates time dependence. The stream function $\Psi(x, y)$, defined by $\Psi_y = u - c$ and $\Psi_x = -v$, is harmonic in Ω and satisfies the free boundary problem

$$\begin{cases} \Delta\Psi = 0 & \text{in } \Omega, \\ \frac{1}{2}|\nabla\Psi|^2 + g(y + d) = Q & \text{on } y = \eta(x), \\ \Psi = 0 & \text{on } y = \eta(x), \\ \Psi = m & \text{on } y = -d, \end{cases} \tag{1}$$

where m and Q are physical constants with $Q > 0$. The complex potential $f = \phi + i\psi$ and the complex velocity $w = u - iv$ are holomorphic functions of $z = x + iy$ throughout Ω.

In plain terms. The wave travels without changing shape. The stream function satisfies Laplace's equation throughout the fluid. That is not a coincidence; it is the reason complex analysis can be brought to bear on this problem at all.

2. Why does the pressure at the bed determine $\mathcal{B}$?

Setup. The Bernoulli equation holds throughout Ω:

$$2p + 2gy + u^2 + v^2 = B$$

for some constant B. At the free surface $y = \eta(x)$ the normalised pressure is $p = 0$. At the bed $y = -d$ it gives the measured quantity $p_b(x) = p(x, -d)$.

Definition 2.1. Let $w(z) = u - iv$ be the complex velocity. Define

$$\mathcal{B}(z) \;=\; \tfrac{1}{2}B + gd - \tfrac{1}{2}w^2(z) \;=\; \tfrac{1}{2}B + gd - \tfrac{1}{2}(u^2 - v^2) + iuv.$$

Proposition 2.2. *At the flat bed $y = -d$,*

$$\mathrm{Im}\big(\mathcal{B}(x - id)\big) = 0 \qquad \textit{and} \qquad \mathrm{Re}\big(\mathcal{B}(x - id)\big) = p_b(x).$$

Proof. At $y = -d$ the bed is impermeable, so $v = 0$. The complex velocity $w(x - id)$ is therefore real, giving $\mathrm{Im}(\mathcal{B}(x - id)) = 0$. For the real part, the Bernoulli equation at the bed gives

$$\mathrm{Re}(\mathcal{B}(x - id)) = \tfrac{1}{2}B + gd - \tfrac{1}{2}u_b^2 = p_b(x),$$

where the last equality uses $2p_b + 2g(-d) + u_b^2 = B$.

In plain terms. The sensor measures $p_b(x)$, which is exactly $\mathrm{Re}(\mathcal{B})$ along the bottom line. Since $\mathcal{B}$ is holomorphic, knowing its real part along a horizontal line determines it everywhere in the fluid by analytic continuation, including at the free surface. That is the step that makes the recovery possible.

Note on the classical procedure. Clamond and Constantin (2012) recover η without constructing $\mathcal{B}$ explicitly. The procedure uses an iterative fixed-point scheme to find the crest amplitude η_0 and an approximation b of B, then integrates a differential equation for the surface. The approach presented here, due to Clamond (2013), bypasses this iteration entirely with a closed-form formula. Both methods are detailed in the complementary questions.

3. What is $\mathcal{D}_s$ and where does it come from?

Setup. Since $\mathcal{B}$ is holomorphic, it can be integrated along any path in the fluid domain.

Definition 3.1. For $z_0 = ia$ (the crest point, where $a = \eta(0)$ is the crest amplitude), define

$$\mathcal{D}(z) = \int_{z_0}^{z} \big(\mathcal{B}(z') - gd\big)\, dz'.$$

Proposition 3.2 (Explicit form of $\mathcal{D}_s$). *Parametrising the free surface as $z(x') = x' + i\eta(x')$ and applying $dz' = (1 + i\eta_{x'})\,dx'$,*

$$\mathcal{D}_s(z) = \int_0^x g\eta(x')\,dx' + i\big[\eta^2(x) - a^2\big]\big[B - \tfrac{1}{2}ga - \tfrac{1}{2}g\eta(x)\big]\,. \tag{2}$$

Proof. At the free surface, $(\mathcal{B}_s - gd)(1 + i\eta_x) = g\eta(1 - i\eta_x) + iB\eta_x$. Expanding and integrating term by term:

$$\begin{aligned}
\mathcal{D}_s(z) &= \int_0^x g\eta\,dx' - ig\int_0^x \eta\eta_x\,dx' + iB\int_0^x \eta_x\,dx' \\
&= \int_0^x g\eta\,dx' - \tfrac{ig}{2}\big[\eta^2\big]_0^x + iB\big[\eta\big]_0^x \\
&= \int_0^x g\eta\,dx' + i\big[\eta^2(x) - a^2\big]\big[B - \tfrac{1}{2}ga - \tfrac{1}{2}g\eta(x)\big]\,.
\end{aligned}$$

In plain terms. $\mathcal{D}_s$ encodes the integral of the surface profile along the wave. Its imaginary part contains η implicitly. That is exactly what we need to extract the wave shape in the next section.

4. How is η extracted from $\mathcal{D}_s$?

Theorem 4.1 (Recovery formula). *The surface elevation η satisfies*

$$\boxed{\eta = g^{-1}\Big[B - \sqrt{(B - ga)^2 - 2g\,\mathrm{Im}(\mathcal{D}_s)}\Big]\,.} \tag{3}$$

Proof. From Proposition 3.2, $\mathrm{Im}(\mathcal{D}_s)$ is a quadratic function of η. Solving for η and retaining the root satisfying $\eta(0) = a$ gives (3).

Corollary 4.2 (Derivative formula). *Differentiating (3) with respect to x:*

$$\eta_x = \frac{[\mathrm{Im}(\mathcal{D}_s)]_x}{\sqrt{(B - ga)^2 - 2g\,\mathrm{Im}(\mathcal{D}_s)}} = \frac{\mathrm{Im}(\mathcal{B}_s) + \big(\mathrm{Re}(\mathcal{B}_s) - gd\big)\eta_x}{B - g\eta}. \tag{4}$$

This provides a local algebraic equation for η_x and confirms that Theorem 4.1 is equivalent to the implicit integral equation derived in the classical procedure.

In plain terms. Once we know $\mathcal{B}$ and $\mathcal{D}_s$, the wave shape follows from a single closed-form expression. No iteration is required. The derivative formula is a consistency check: it says the slope of the recovered wave at every point agrees with what $\mathcal{B}$ predicts directly.

5. How is B determined?

B appears throughout the formulas above but is not directly measurable. Two methods are available.

Method 1 (Non-local). Taking the spatial average of the imaginary part of Corollary 4.2 and using $\langle\eta\rangle = 0$:

$$2aB = g\big(a^2 - \langle\eta^2\rangle\big) - 2\,\mathrm{Im}(\mathcal{D}_s).$$

This relation is non-local because it involves integration over a full wavelength.

Method 2 (Implicit equations, used in the code). Let a and b' be the solutions to

$$a = g^{-1}\big(\mathrm{Re}(\mathcal{B}(ia)) - d\big), \tag{5}$$

$$b' = d - g^{-1}\,\mathrm{Re}\big(\mathcal{B}(\pi/k - ib')\big). \tag{6}$$

Then

$$B = \tfrac{1}{2}g(a - b') - (a + b')^{-1}\,\mathrm{Im}\big(\mathcal{D}(\pi/k - ib')\big). \tag{7}$$

This is the method implemented in `wave_recovery.py`. The two scalar equations for a and b' are solved numerically with `fsolve`; B then follows directly from (7).

> **In plain terms.** a is the height of the crest above the mean level; b' is the depth of the trough below it. Both can be read from $\mathcal{B}$ at two specific complex points, the crest and trough positions in the complex plane. From those two numbers, B follows algebraically.

6. What breaks at high N?

Setup. To use the formulas numerically, we approximate $p_b(x)$ as a Fourier series of order N with coefficients p_n obtained from the 1024 pressure measurements:

$$p(x) = gd + \sum_{\substack{n=-N \\ n\neq 0}}^{N} p_n \exp(inkx).$$

This gives explicit formulas for $\mathcal{B}$ and $\mathcal{D}$:

$$\mathcal{B}(z) = gd + \sum_{|n|>0}^{N} p_n \frac{\exp(inkz)}{\exp(nkd)}, \tag{8}$$

$$\mathcal{D}(z) = \sum_{|n|>0}^{N} \frac{p_n}{ink} \cdot \frac{\exp(inkz) - \exp(-nka)}{\exp(nkd)}. \tag{9}$$

Proposition 6.1 (Amplification factor)**.** *For mode n, the reconstruction multiplies the measured Fourier coefficient p_n by $\exp(nkd)$. For the dataset used here ($kd \approx 1.34$):*

n	$\exp(nkd)$
1	3.8
5	7.6×10^2
10	5.8×10^5
20	3.4×10^{11}

The measured coefficient carries a rounding error $|\delta_n|$ of order machine precision. When $\exp(nkd) \cdot |\delta_n|$ exceeds $|p_n|$, the amplified error dominates the signal.

Numerical evidence. Running the recovery at increasing N on the thesis dataset:

N	Δ_1 (crest error)	Δ_2 (trough error)
5	2.6×10^{-4}	1.4×10^{-5}
10	7.3×10^{-7}	1.3×10^{-9}
20	6.0×10^{-5}	2.6×10^{-9}
40	4.5×10^{-1}	1.1×10^{-1}
100	6.5×10^{-1}	1.3

The error improves up to $N \approx 10$, then deteriorates. At $N = 40$ the crest error exceeds the wave amplitude itself.

In plain terms. The data is faithful. The algorithm is correct. What fails is the inversion of the water column's attenuation at high frequency. The optimal N is not a property of the data; it is a property of the physics. Past the threshold where $\exp(nkd)$ amplifies noise above the signal, more Fourier terms make things strictly worse.

More precision is not always more truth. Beyond a threshold, the model stops listening to the data and starts listening to itself.

Complementary questions

These sections expand on background material used implicitly in the main argument.

A. What are the physical hypotheses?

The model assumes a two-dimensional periodic wave, irrotational and inviscid flow, a flat impermeable bed at $y = -d$, and a free surface at $y = \eta(x)$ where the normalised pressure vanishes. The flow is steady in the moving frame. Mass conservation gives $u_x + v_y = 0$; the Euler equations reduce to the Bernoulli condition $2p + 2gy + u^2 + v^2 = B$. The mean pressure at the bed satisfies $\langle p_b \rangle = gd$. These hypotheses are developed in Chapter 1 of the original manuscript.

B. What inequalities does the pressure satisfy?

Two results from Chapter 3 are used implicitly throughout.

- For all $y \in [-d, \eta_0]$, $p_y(0, y) < 0$: the pressure is strictly decreasing with height along the crest line, and attains its minimum only at the free surface where it is zero.
- In the moving frame, $p_x < 0$ in Ω_+ and $p_x > 0$ in Ω_-, except on the crest and trough lines.

Both results are proved using Hopf's maximum principle applied to the superharmonic function P in the fluid domain.

C. What is the classical procedure?

Clamond and Constantin (2012) recover η in three steps.

Step 1. The crest amplitude η_0 is found as the unique fixed point of $F(y) = y + p(0, y)/g$, shown to be a contraction on $[-d, \eta_0]$.

Step 2. The Bernoulli constant B is approximated by a value b satisfying $b \approx g(\eta_0 - d) + \mathrm{Re}(\mathcal{B}(0, \eta_0))$.

Step 3. The surface is recovered by iterating

$$\eta_{n+1}(x) = \eta_0 + \int_0^x \frac{\mathrm{Im}(\mathcal{B}(x', \eta_n(x')))}{b - g\eta_n - \mathrm{Re}(\mathcal{B}(x', \eta_n)) + gd}\, dx'$$

until convergence. The Picard-Lipschitz theorem guarantees uniform convergence on $[0, L/2]$.

D. Why does the Banach fixed point theorem apply?

In Step 1 above, $F(y) = y + p(0, y)/g$ is shown to be a contraction on $[-d, \eta_0]$. Working at the crest line $x = 0$ where $v = 0$:

$$|F(y) - F(y')| = \left| y - y' + \frac{p(0, y) - p(0, y')}{g} \right| = \frac{1}{2g} \left| u(0, y') - u(0, y) \right| \leq \frac{1}{2g} \left| y' - y \right|,$$

where the last inequality uses the horizontal velocity bound from Chapter 3. Since $1/(2g) < 1$ in normalised units ($g = 1$), F is a contraction and Banach's theorem gives a unique fixed point η_0, obtained as the limit of $y_n = F(y_{n-1})$.

References

[1] D. Clamond, *New exact relations for easy recovery of steady wave profiles from bottom pressure measurements*, 2013.

[2] D. Clamond and A. Constantin, *Recovery of steady periodic wave profiles from pressure measurements at the bed*, Cambridge University Press, 2012.

[3] A. Constantin and W. Strauss, *Pressure beneath a Stokes wave*, Wiley Periodicals, 2009.